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Patent Public Search | Text View

United States Patent Application Publication

20250281263

Kind Code

A1

Publication Date

September 11, 2025

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Orthodontic Gripping Device

Abstract

The present invention relates to an orthodontic gripping device having a body made of a superelastic metal alloy, cold worked titanium beta III, or solution heat treated and aged titanium beta III. The body has first and second, spaced apart arm portions connected to each other. Each arm portion includes a jaw portion having an inner, arch wire-gripping, surface. The inner gripping surfaces are disposed opposite one another in spaced relation with a station defined therebetween for receiving the arch wire. At least a portion of the station is sized smaller than the arch wire. The arm portions are resiliently deflectable outwardly away from each other to admit the arch wire into the station. Once seated in the station, the arch wire is tightly held by inner gripping surfaces so as to resist displacement of the gripping device relative to the arch wire.

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Appl. No.: 19/216941

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18390975 20231220 PENDING child US 19216941

parent US continuation 17695437 20220315 parent-grant-document US 11890161 child US 18390975

parent US continuation 13373968 20111207 parent-grant-document US 9877804 child US 15852344

parent US division 15852344 20171222 parent-grant-document US 11304780 child US 17695437
us-provisional-application US 61457013 20101208

Publication Classification

Int. Cl.: **A61C7/02** (20060101); **A61C7/04** (20060101); **A61C7/20** (20060101); **A61C7/22** (20060101)

U.S. Cl.:

CPC **A61C7/02** (20130101); **A61C7/04** (20130101); **A61C7/20** (20130101); **A61C7/22** (20130101);

Background/Summary

RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/390,975, filed on Dec. 20, 2023. U.S. patent application Ser. No. 18/390,975, is a continuation application of U.S. patent application Ser. No. 17/695,437 filed on Mar. 15, 2022 and issued as U.S. Pat. No. 11,890,161 on Feb. 6, 2024. U.S. patent application Ser. No. 17/695,437 is a divisional application of U.S. patent application Ser. No. 15/852,344, filed on Dec. 22, 2017 and issued as U.S. Pat. No. 11,304,780 on Apr. 19, 2022. U.S. patent application Ser. No. 15/852,344 is a continuation application of U.S. patent application Ser. No. 13/373,968 filed on Dec. 7, 2011, which issued as U.S. Pat. No. 9,877,804 and which claims priority under 35 U.S.C. § 120 to U.S. Provisional Patent Application Ser. No. 61/457,013 filed on Dec. 8, 2010, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of orthodontics, in particular, to an orthodontic gripping device suitable for tightly clamping onto an arch wire or other orthodontic component.

BACKGROUND OF THE INVENTION

[0003] In orthodontics, an arch wire is used in conjunction with orthodontic brackets mounted to the patient's teeth to move the teeth to desired positions. The arch wire is configured to conform generally to the dental arch of the patient and is accommodated or seated within arch wire slots defined in the orthodontic brackets. The arch wire exerts forces on the orthodontic brackets which over time cause the teeth to move to their correct positions.

[0004] In orthodontic treatment, it is customary to place arch wire stops at specific locations on the arch wire to prevent displacement of the arch wire in the mesial/distal direction relative to one or more brackets. In some instances, arch wire stops will be provided to arrest movement at the free ends of the arch wire to prevent the arch wire from disengaging from the arch wire slot of the distalmost bracket and from making contact with (and possibly, irritating) the soft tissue (e.g. gums) in the patient's mouth. In other instances, stops may be used to fix the arch wire to a bracket/tooth combination to effect a particular transmission of forces to the bracket/tooth combination to achieve a particular result (e.g. to prevent migration of a tooth along the arch wire).

[0005] The arch wire stops that have been developed to date use a variety of techniques and/or structural components to limit mesial/distal displacement of the arch wire relative to a bracket. One known design employs a generally C-shaped clamp provided with an arch wire receiving slot. A closure member is pivotally connected to the clamp and can be moved between an open position to allow access to the arch wire receiving slot and a closed position wherein access to the arch wire receiving is restricted. The closure member has a variable thickness and is formed with an inner camming surface for engagement with the arch wire. When the closure member is moved to the open position, the arch wire receiving slot remains unobstructed and can receive an arch wire. When the closure member is moved to the closed position, the inner camming surface is brought to

bear against the outer surface of the arch wire and creates a mechanical constriction (or narrowing) of the arch wire slot. As a result, the arch wire is retained between the inner walls of the clamp and the inner camming surface, and is discouraged from moving mesially or distally relative to a bracket by the friction forces acting on the arch wire.

[0006] Still other stops use different movable components to tightly hold the arch wire within the stop. For example, in some stops, a rotatable screw is threadingly engaged with a sleeve so as to project into the arch wire slot defined in the sleeve. As the screw is tightened, its lingual end bears against the outer surface of the arch wire trapping it within the sleeve.

[0007] The relatively small size of these types of stops and the fact that they include movable components have tended to make them difficult to handle and deploy in the field. Moreover, there have been some complaints that once mounted in the mouth these stops tend to be bulky and may irritate the patient's cheek or gum tissue as a result of their bulkiness. The design of these stops may also allow food to become trapped within the stop or between the stop and the patient's soft tissue or dentition. Additionally, these stops tend to loosen from, or slip along, the arch wire over time.

[0008] A more common approach to restricting mesial/distal displacement of an arch wire involves the use of crimpable, tubular or C-shaped, sleeves. The arch wire is inserted through the sleeve and its position is fixed relative to the arch wire by crimping the sleeve in place. Slippage problems have been encountered with these types of stops, as the arch wire tends to slide within the sleeve when forces are applied to it. To address these concerns, some sleeves have been provided with roughened or irregular inner wire-engaging surfaces to increase friction and enhance the gripping action of the sleeve. These roughened surfaces may be obtained by coating the inner wire-engaging surfaces with abrasive material or by mechanically roughening the surfaces. These modifications have tended to lessen the occurrence of slippage of the arch wire. However, handling and slippage problems remain. Threading of the arch wire through the sleeve can be challenging as the arch wire itself has a relatively small cross-sectional area and the passageway formed in the sleeve is also quite small. Once threaded on the arch wire (and prior to crimping) the sleeve can inadvertently move during crimping thus making precise positioning difficult.

[0009] In light of the foregoing, it would be advantageous to have an orthodontic gripping device or arch wire stop with enhanced gripping action to overcome the slippage problems heretofore experienced with existing arch wire stops. Preferably, such an arch wire gripping device would be relatively easy to handle and quick to attach on an arch wire.

SUMMARY OF THE INVENTION

[0010] According to a broad aspect of an embodiment of the present invention, there is provided an orthodontic gripping device for attaching to an arch wire. The gripping device includes a body made of a material selected from the group consisting of a superelastic metal alloy, cold worked titanium beta III, and solution heat treated and aged titanium beta III. The body has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner, arch wire-gripping surface. The second arm portion includes a second jaw portion having a second inner, arch wire-gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the arch wire therein. At least a portion of the arch wire receiving station is sized smaller than the arch wire. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the arch wire receiving station. The first and second arm portions are resiliently deflectable outwardly away from each other when the arch wire is admitted through the entranceway into the arch wire receiving station. The first and second inner gripping surfaces engage the arch wire and apply opposing gripping forces against the arch wire so as to resist displacement of the gripping device relative to the arch wire, when the arch wire is seated within the arch wire receiving station.

[0011] In another feature, the body is of unitary construction. In a further alternative, at least one of

the arm portions is a separate component. Optionally, the superelastic metal alloy may be selected from the group consisting of: (a) cold worked nickel titanium; (b) cold worked and aged nickel titanium; (c) cold worked nickel titanium and other alloying elements; and (d) cold worked and aged nickel titanium and other alloying elements. In one feature, the superelastic metal alloy behaves in a linear superelastic mode. In another feature, the superelastic metal alloy behaves in a non-linear superelastic mode. In alternative feature, the body is made of a material consisting of: (a) cold worked titanium beta III; and (b) solution heat treated and aged titanium beta III. [0012] In an additional feature, the body has a longitudinal centerline. The first and second arm portions are disposed on opposite sides of the longitudinal centerline and are mirror images of each other.

[0013] In yet another feature, the body has a longitudinal centerline and the body is symmetrical about the longitudinal centerline. Alternatively, the body may be asymmetrical about the longitudinal centerline. In still another feature, the body has a transverse centerline and is symmetrical about the transverse centerline. In a further alternative feature, the body may be asymmetrical about the transverse centerline.

[0014] In one feature, the body further includes a base portion extending between the first and second arm portions to connect one to the other. The entranceway is disposed opposite the base portion.

[0015] In another feature, the arch wire has a profile defined by an outer surface. The first and second gripping surfaces are configured to substantially conform to at least a portion of the arch wire's outer surface. Additionally, the profile of the arch wire is generally circular. Each of the first and second inner gripping surfaces has an arcuate profile configured to at least partially match the curvature of the arch wire's profile. In a further feature, the arch wire has a diameter. The arch wire receiving station has a substantially circular profile and a diameter. The diameter of the arch wire receiving station is sized smaller than the diameter of the arch wire.

[0016] In yet another feature, the arch wire has a width. At least a portion of the arch wire receiving station is sized smaller than the width of the arch wire. Optionally, the arch wire receiving station may have a width of constant dimension. The width of the arch wire receiving station could be sized smaller than the width of the arch wire. In addition, the body has a longitudinal centerline. The first and second arm portions are disposed on opposite sides of the longitudinal centerline. The first and second arm portions are mirror images of each other. The first and second inner gripping surfaces are oriented substantially parallel to the longitudinal centerline.

[0017] In an optional feature, the arch wire has a width. The arch wire receiving station has a width of variable dimension. The smallest width of the arch wire receiving station is sized smaller than the width of the arch wire. The body has a longitudinal centerline. The first and second arm portions are disposed on opposite sides of the longitudinal centerline and are mirror images of each other. The first and second inner gripping surfaces have a skewed orientation relative to the longitudinal centerline. Each gripping surface converges toward the other in the direction of the entranceway.

[0018] In an additional feature, each inner gripping surface is a surface selected from the group consisting of: (a) a smooth surface; (b) an irregular surface; (c) a textured surface; and (d) a surface coated with a friction enhancing material.

[0019] In a further feature, the entranceway is funnel-shaped and tapers in the direction of the arch wire receiving station. The entranceway terminates in a throat and the throat is sized smaller than the size of the arch wire. In one optional feature, the throat is sized between 35% and 40% of the diameter of the arch wire. In another optional feature, the throat is sized between 65% and 70% of the width of the arch wire.

[0020] In still another feature, the gripping device further includes an orthodontic accessory formed in the body. The orthodontic accessory is selected from the group consisting of: (a) a hook; (b) an auxiliary slot; and (c) an eyelet. In another feature, the gripping device further includes a hook

extending from the body. Optionally, the hook may be integrally formed with the body. In another feature, the hook is carried on the first arm portion. The first arm portion has an outer surface. The hook may extend perpendicular to the outer surface of the first arm portion, or alternatively, the hook may extend from the outer surface of the first arm portion on a slant.

[0021] In yet another feature, the body further includes a base portion extending between the first and second arm portions to connect one to the other. The hook is carried on the base portion. The base portion has an outer surface. The hook may extend perpendicular to the outer surface of the base portion or alternatively, the hook may extend from the outer surface of the base portion on a slant.

[0022] In still another feature, the body further includes an outer surface and the hook extends from the outer surface of the body.

[0023] In a further feature, the arch wire receiving station is shaped to accommodate an arch wire having a profile selected from the group consisting of: (a) a circular profile; (b) a rectangular profile; (c) a square profile; and (d) a D-shaped profile.

[0024] In yet another feature, the gripping device is provided with means for facilitating handling of the gripping device. The handling means is formed on the body. More specifically, the handling means includes a first groove defined in the outer surface of the first arm portion and a second groove defined in the outer surface of the second arm portion. The first groove is disposed opposite the second groove. In another feature, the handling facilitating means includes first and second lateral wing members extending from the base portion. the first lateral wing member is disposed in spaced relation relative to the first arm portion. The second lateral wing member is disposed in spaced relation relative to the second arm portion. In a further feature, each of the first and second lateral wing members has an outer lateral surface and is moveable between an undeflected position and a deflected position. The gripping device has a width measured between the outer lateral surface of the first lateral wing member and the outer lateral surface of the second lateral wing member. The width of the gripping device when the first and second lateral wing members are in the deflected position is smaller than the width of the gripping device when the first and second lateral wing members are in the undeflected position.

[0025] In an additional feature, the arch wire receiving station has a profile selected from the group consisting of: (a) a circular profile; (b) an oval profile; (c) a rectangular profile; (d) a square profile; and (e) a quatrefoil profile.

[0026] In one feature, the body has a longitudinal centerline. The arch wire receiving station has a substantially oval profile defined by a major axis and a minor axis. The minor axis is disposed parallel to the longitudinal centerline. In another feature, the arch wire has a diameter and the arch wire receiving station has a diameter measured along the minor axis. The diameter of the arch wire receiving station measured along the minor axis is sized smaller than the diameter of the arch wire.

[0027] According to a broad aspect of another embodiment of the present invention, there is provided an orthodontic gripping device for attaching to an arch wire. The gripping device includes a body made of a shape memory material having a shape reset temperature range within which the shape memory material is capable of deforming significantly from a memorized shape of the shape memory material and a shape recovery temperature range within which the shape memory material is capable of recovering the memorized shape. The body has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner, arch wire-gripping surface. The second arm portion includes a second jaw portion having a second inner, arch wire-gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the arch wire therein. At least a portion of the arch wire receiving station is sized smaller than the arch wire. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the arch wire receiving station. The first and second arm portions are movable outwardly away from each

other when the temperature of the shape memory material of the body lies within the shape reset temperature range and remain spread apart as long as the temperature of the shape memory material lies below the shape recovery temperature range, so as to allow admission of the arch wire through the entranceway into the arch wire receiving station. The first and second inner gripping surfaces engage the arch wire and apply opposing gripping forces against the arch wire so as to resist displacement of the gripping device relative to the arch wire, when the arch wire is seated within the arch wire receiving station and the temperature of the shape memory material of the body lies within the shape recovery temperature range.

[0028] In an additional feature, the shape memory material is selected from the group consisting of: (a) a shape memory metal alloy; and (b) a shape memory non-metallic material. In another feature, the shape recovery temperature range of the shape memory material lies below the temperature of a patient's oral cavity. In further feature, the shape recovery temperature range of the shape memory material encompasses the temperature of a patient's oral cavity. In an alternate feature, the shape recovery temperature range of the shape memory material is above the temperature of a patient's oral cavity and the shape reset temperature range of the shape memory material lies below the temperature of a patient's oral cavity.

[0029] According to a broad aspect of another embodiment of the present invention, there is provided an orthodontic gripping device for attaching to an arch wire. The gripping device includes a body which has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner, arch wire-gripping surface. The second arm portion includes a second jaw portion having a second inner, arch wire-gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the arch wire therein. At least a portion of the arch wire receiving station is sized smaller than the arch wire. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the arch wire receiving station. The first and second arm portions are resiliently deflectable outwardly away from each other when the arch wire is admitted through the entranceway into the arch wire receiving station. The first and second inner gripping surfaces engage the arch wire and apply opposing gripping forces against the arch wire so as to resist displacement of the gripping device relative to the arch wire, when the arch wire is seated within the arch wire receiving station. In a further feature, the body is made of a resilient spring-back material.

[0030] According to a broad aspect of still another embodiment of the present invention, there is provided an orthodontic arch wire kit. The kit includes an arch wire and a gripping device for attaching to an arch wire. The gripping device includes a body made of a material selected from the group consisting of a superelastic metal alloy, cold worked titanium beta III, and solution heat treated and aged titanium beta III. The body has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner, arch wire-gripping surface. The second arm portion includes a second jaw portion having a second inner, arch wire-gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the arch wire therein. At least a portion of the arch wire receiving station is sized smaller than the arch wire. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the arch wire receiving station. The first and second arm portions are resiliently deflectable outwardly away from each other when the arch wire is admitted through the entranceway into the arch wire receiving station. The first and second inner gripping surfaces engage the arch wire and apply opposing gripping forces against the arch wire so as to resist displacement of the gripping device relative to the arch wire, when the arch wire is seated within the arch wire receiving station.

[0031] In another feature, the arch wire has a cross-sectional profile selected from the group

consisting from: (a) circular; (b) rectangular; (c) square; and (d) D-shaped. In one optional feature, the arch wire is selected from the group consisting of: (a) braided wire strands; (b) helically wrapped strands. Additionally, the arch wire could be selected from the group consisting of: (a) a multi-strand arch wire with a core; and (b) a multi-strand coreless arch wire. In another optional feature, the arch wire is a single solid wire.

[0032] In still another feature, the arch wire is made of material selected from the group consisting of: (a) a linear elastic material; (b) a superelastic material; and (c) a shape memory material.

[0033] According to a broad aspect of yet another embodiment of the present invention, there is provided a dental gripping device for attaching to a dental component. The gripping device includes a body made of a material selected from the group consisting of a superelastic metal alloy, cold worked titanium beta III, and solution heat treated and aged titanium beta III. The body has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner gripping surface. The second arm portion includes a second jaw portion having a second inner gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the dental component. At least a portion of the receiving station is sized smaller than the dental component. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the receiving station. The first and second arm portions are resiliently deflectable outwardly away from each other when the dental component is admitted through the entranceway into the receiving station. The first and second inner gripping surfaces engages the dental component and applies opposing gripping forces against the dental component so as to resist displacement of the gripping device relative to the dental component, when the dental component is seated within the receiving station.

[0034] According to a broad aspect of still another embodiment of the present invention, there is provided a gripping device for attaching to an object to be gripped. The gripping device includes a body made of a material selected from the group consisting of a superelastic material, cold worked titanium beta III, and solution heat treated and aged titanium beta III. The body has first and second, spaced apart arm portions connected to each other. The first arm portion includes a first jaw portion having a first inner gripping surface. The second arm portion includes a second jaw portion having a second inner gripping surface. The first and second inner gripping surfaces are disposed opposite one another in spaced relation. The space between the first and second inner gripping surfaces defines a station for receiving the object to be gripped. At least a portion of the receiving station is sized smaller than the object to be gripped. The body further includes an entranceway defined between the first and second arm portions. The entranceway provides access to the receiving station. The first and second arm portions are resiliently deflectable outwardly away from each other when the object to be gripped is admitted through the entranceway into the receiving station. The first and second inner gripping surfaces engages the object to be gripped and applies opposing gripping forces against the object to be gripped so as to resist displacement of the gripping device relative to the object to be gripped, when the object to be gripped is seated within the receiving station.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The embodiments of the present invention shall be more clearly understood with reference to the following detailed description of the embodiments of the invention taken in conjunction with the accompanying drawings, in which:

[0036] FIG. 1 is a perspective view of an orthodontic gripping device in accordance with a first

embodiment of the invention;

[0037] FIG. 2A is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. 1;

[0038] FIG. 2B is a side elevation view of the orthodontic gripping device shown in FIG. 1;

[0039] FIG. 3 is another perspective view showing the orthodontic gripping device of FIG. 1 attached to an arch wire of circular or substantially circular cross-section;

[0040] FIG. 4 is an end elevation (or mesial) view of the arch wire and orthodontic gripping device shown in FIG. 3;

[0041] FIG. 5A is a perspective view of an orthodontic gripping device in accordance with a second embodiment of the invention showing an orthodontic gripping device provided with a hook;

[0042] FIG. 5B is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. 5A;

[0043] FIG. 5C is a side elevation view of the orthodontic gripping device shown in FIG. 5A;

[0044] FIG. 6 is a perspective view of an orthodontic pair of pliers in accordance with an embodiment of the invention used to attach a gripping device onto an orthodontic arch wire;

[0045] FIG. 7 is a side elevation view of the pliers shown in FIG. 6;

[0046] FIG. 8 is an enlarged perspective view of the first male and second female work tips of the pliers illustrated in FIG. 6, showing the pliers in an open setting with the male work tip spaced apart from the female work tip;

[0047] FIG. 9 is an enlarged, isolated perspective view of the female work tip shown in FIG. 8;

[0048] FIG. 10 is an end elevation view looking directly at the work tips of the pliers illustrated in FIG. 6 with a portion of the work tips removed for clarity, showing a gripping device of the type shown in FIG. 1 seated in a rebate defined in the female work tip of the pliers and the flange of the male work tip bearing against the outer surface of the arch wire;

[0049] FIG. 11 is magnified elevation view of the work tips and gripping device illustrated in FIG. 10, with the arm portions of the gripping device shown deflecting outwardly from each other as the arch wire is admitted through the entranceway of the gripping device body;

[0050] FIG. 12 is a perspective view of an orthodontic gripping device in accordance with a third embodiment of the invention;

[0051] FIG. 13A is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. 12;

[0052] FIG. 13B is a side elevation view of the orthodontic gripping device shown in FIG. 12;

[0053] FIG. 14 is another perspective view showing the orthodontic gripping device of FIG. 12 attached to an arch wire with a substantially rectangular cross-section;

[0054] FIG. 15 is an end elevation (or mesial) view of the arch wire and orthodontic gripping device shown in FIG. 14;

[0055] FIG. 16 is an end elevation view looking directly at the work tips of the pliers illustrated in FIG. 6, showing a gripping device of the type shown in FIG. 12 seated in a rebate defined in the female work tip of the pliers and the flange of the male work tip bearing against the outer surface of the arch wire;

[0056] FIG. 17 is magnified elevation view of the work tips and gripping device illustrated in FIG. 16 with a portion of the work tips removed for clarity, with the arm portions of the gripping device shown deflecting outwardly from each other as the arch wire is admitted through the entranceway of the gripping device body;

[0057] FIG. 18 is a perspective view of an orthodontic gripping device in accordance with a fourth embodiment of the invention;

[0058] FIG. 19 is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. 18;

[0059] FIG. 20 is a perspective view of an orthodontic gripping device in accordance with a fifth embodiment of the invention; and

[0060] FIG. **21** is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. **20**;

[0061] FIG. **22** is a perspective view of an orthodontic gripping device in accordance with a sixth embodiment of the invention; and

[0062] FIG. **23** is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. **22**;

[0063] FIG. **24A** is a magnified elevation view similar to that illustrated in FIG. **11** showing the female work tip only of the orthodontic pliers and the gripping device of FIG. **23** with the lateral wing members of the gripping device occupying their respective undeflected positions, prior to insertion of the gripping device into the rectangular rebate of the female work tip;

[0064] FIG. **24B** is a magnified elevation view of the female work tip and the gripping device illustrated in FIG. **24A** showing the lateral wing members of the gripping device moved to their respective deflected positions thereby allowing the gripping device to fit within the rectangular rebate of the female tip portion;

[0065] FIG. **25** is a perspective view of an orthodontic gripping device in accordance with a seventh embodiment of the invention; and

[0066] FIG. **26** is an end elevation (or mesial) view of the orthodontic gripping device shown in FIG. **25**.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

[0067] The description, which follows, and the embodiments described therein are provided by way of illustration of an example, or examples of particular embodiments of principles and aspects of the present invention. These examples are provided for the purposes of explanation and not of limitation, of those principles of the invention. In the description that follows, like parts are marked throughout the specification and the drawings with the same respective reference numerals.

[0068] As used in this specification, the term “superelastic material” is understood to mean a material acting or behaving in a linear superelastic mode or in a non-linear superelastic mode. Similarly, a “superelastic metal alloy” means a metal alloy acting or behaving in a linear superelastic mode or in a non-linear superelastic mode. In the context of this specification, a material is considered to behave in a linear superelastic mode when it is able to undergo strains approximately 4%, without incurring excessive permanent set after unloading, all the time exhibiting a substantially linear relationship between stress and strain. A material is considered to behave in a non-linear superelastic mode when it exhibits the ability to incur relatively large amounts of strain without incurring excessive permanent set after unloading, and the loading and unloading stress-strain curves for the material are non-linear or multi-linear for the majority of their extent.

[0069] The term “shape memory material” is understood to mean a material acting or behaving in a shape memory mode. In like fashion, a “shape memory metal alloy” means a metal alloy acting or behaving in a shape memory mode. In the context of this specification, a material is considered to behave in a shape memory mode when it has the ability to both undergo deformation of a previously memorized shape and subsequently recover that memorized shape when exposed to some type of external stimulus, most commonly a reversible temperature change (as is the case, when the material is a metal alloy).

[0070] Referring to FIGS. **1**, **2A** and **2B**, there is shown an orthodontic gripping device in accordance with a first embodiment of the present invention designated generally with reference numeral **20**. The gripping device **20** is configured for use with an orthodontic arch wire. The orthodontic gripping device **20** includes a mesial face **22**, a distal face **24** and an elongate body **26** extending between the mesial and distal faces **22** and **24**. In this embodiment, the body **26** is of unitary (one-piece or monolithic) construction fabricated from a superelastic material. Preferably, such material is a superelastic metal alloy. Most preferably, the superelastic metal alloy is nickel titanium, or nickel titanium and other alloying elements (e.g. chromium (Cr), iron (Fe), vanadium

(V), aluminium (Al), copper (Cu), cobalt (Co)), such as nickel/titanium/copper or nickel/titanium/copper/chromium. Preferably, the nickel titanium, or nickel titanium and other alloying elements, is cold worked. Although, cold worked and aged nickel titanium or cold worked and aged nickel titanium and other alloying elements, could also be used. In this embodiment, the gripping device **20** is made of cold worked nickel titanium.

[0071] However, it should be appreciated that the gripping device could also be made from other materials suitable for use in orthodontic appliances and exhibiting spring-back properties similar to those exhibited by superelastic materials. For example, a gripping device fabricated from cold worked titanium beta III or solution heat treated and aged titanium beta III (a titanium molybdenum alloy) has been found to have superior gripping strength. Testing has shown that the gripping device could also be made of cold worked ELGILOY™ alloy (a cobalt, chromium and nickel alloy). Still other materials may be selected based on their ability to resist permanent deformation and/or mechanical failure when the gripping device is flexed during its engagement with the arch-wire, while at the same time possessing sufficient material stiffness, such that the gripping device exhibits sufficient spring-back to hold the arch-wire firmly in place once the gripping device is fully engaged with the arch wire.

[0072] In still other embodiments, the gripping device could be made of a shape memory material, such a shape memory metal alloy. Although, conceivably, a non-metallic material (such as a polymer) possessing shape memory properties could also be used.

[0073] For reasons of strength and robustness, it is generally preferred that the body be made monolithic. However, in certain applications it may be desirable to have the body fabricated from separate components. These components could all be made of the same superelastic or shape memory material. Alternatively, the body could be made of a composite of superelastic materials or a composite of shape memory materials. In a further alternative embodiment, the body could be made of a composite of one or superelastic materials with a non-superelastic material or a composite of one or more shape memory materials with a non-shape memory material.

[0074] Preferably, the orthodontic gripping device is manufactured using material removal techniques. However, it is possible the gripping device could also be fabricated using other manufacturing techniques, such as rapid prototyping, sintering, casting, extrusion etc. In cases, where the body is made of separate components, each of these components could be made using one of the afore-mentioned methods or the like and subsequently assembled to each other to form the body by laser welding, fusing, or other similar process.

[0075] The body **26** has a vaguely rectangular shape or profile when seen in a mesial view such as shown in FIG. 2A. The shape of the body **26** is defined substantially by four smoothly radiused corners-first corner **30**, second corner **32**, third corner **34** and fourth corner **36**. The corners **30**, **32**, **34** and **36** are rounded so as to avoid the formation of abrupt projecting edges which could otherwise irritate the soft tissue in the patient's mouth and cause discomfort when the orthodontic gripping device **20** is deployed in the patient's mouth.

[0076] The body **26** includes a longitudinal centreline CL.sub.1 (shown in FIG. 2A), a base portion **38** and a pair of opposed, spaced apart, first and second arm portions **40** and **42** connected to the base portion **38** and disposed on opposite sides of the longitudinal centerline CL.sub.1. The body **26** is symmetrical about the longitudinal centerline CL.sub.1 such that the arm portions **40** and **42** are mirror images of each other. The body **26** is also configured to be symmetrical about a transverse centerline CL.sub.2 (shown in FIG. 2B). This need not be the case in every application. In other embodiments, the body could be configured asymmetrically about one or both the longitudinal centerline CL.sub.1 and the transverse centerline CL.sub.2.

[0077] The first arm portion **40** joins the base portion **38** at the first corner **30**, while the second arm portion **42** attaches to the base portion **38** at the second corner **32**. Each arm portion **40**, **42** extends away from corner **30**, **32**, respectively, in a direction generally perpendicular to the base portion **38**. At the third corner **34**, the first arm portion **40** turns inwardly toward the second arm

portion **42** and ultimately, terminates with a first arch wire engaging jaw portion **44**. Similarly, the second arm portion **42** turns inwardly toward the first arm portion **40** at the fourth corner **36** to terminate with a second arch wire engaging jaw portion **46**.

[0078] As shown in FIG. 2A, the first and second jaw portions **44** and **46** are arranged opposite one another and are separated by a relatively small gap which defines an arch wire receiving slot or station **48** for captively engaging (securely retaining) an arch wire **50** therein. Configured in this manner, the first and second jaw portions **44** and **46** together define the gripping jaw **52** of the orthodontic gripping device **20**.

[0079] While in the preferred embodiment, the body **26** is formed with two opposed arm portions **40** and **42** joined to and integrally formed with the base portion **38**, in other embodiments, the body could be configured differently. For instance, one or both of the arm portions could be fabricated as separate components that are subsequently fixed to the base portion.

[0080] In another embodiment, the body could have two pairs of arms portions mounted in opposition to each other. In such an embodiment, a first pair of arm portions would include first and second, spaced apart, arm portions spaced side-by-side. Similarly, the second pair of arm portions would include third and fourth, spaced apart, arm portions spaced side-by-side. The first arm portion would be disposed opposite the third arm portion, while the second arm portion would be arranged opposite the fourth arm portion. Each of these arm portions would possess an arch wire engaging jaw portion. Defined between the jaw portions of the first and second arm portions, and the jaw portions of the third and fourth arm portions, would be arch wire receiving slots or stations for captively engaging (securely retaining) an arch wire therein. Configured in this manner, the jaw portions of the first and second arm portions together would define a first gripping jaw, while the jaw portions of the third and fourth arm portions together would define a second gripping jaw. In still other embodiments, more than two pairs of opposed arm portions may be provided. Moreover, the body could also be configured so that two or more, smaller arm portions are mounted in opposition to a single, larger arm portion.

[0081] Still other modifications may be brought to the body. In the embodiment shown in FIG. 2A, the body **26** is formed with a well-defined base portion **38**. In other embodiments, the base portion could be configured to be less prominent a structure. In yet other embodiments, the base portion could be omitted altogether. In such embodiments, the arm portions would connect to each other.

[0082] With reference to FIGS. 2A and 2B, the elongate body **26** further includes a length $L_{sub.1}$ measured between the mesial and distal faces **22** and **24** of the body **26**, a width $W_{sub.1}$ measured between the outer lateral surface **60** of the first arm portion **40** and the outer lateral surface **62** of the second arm portion **42**, and a depth $H_{sub.1}$ measured between the outer surface **64** of the base portion **38** and the outer surfaces **66** of the arm portions **40** and **42** disposed opposite the outer surface **64**. In this embodiment, the body **26** is nearly as long as it is wide. In other embodiments, the body may be configured to have a length $L_{sub.1}$ equal to, greater than or lesser than the width $W_{sub.1}$.

[0083] Roughly midway between the third and fourth corners **34** and **36** and opposite the base portion **38**, an arch wire entranceway **70** is formed within the body **26**. The entranceway **70** runs the entire length of the body **26** and is generally funnel-shaped with a taper in the direction of the base portion **38** ultimately culminating in a throat **72**. The funnel shape of the entranceway **70** is defined by a pair of opposed, first and second inclined body surfaces **74** and **76**. The inclined body surfaces **74** and **76** are separated from each other by an angle $\theta_{sub.1}$. In this embodiment, the angle $\theta_{sub.1}$ measures approximately 60.8 degrees. It will be appreciated that in other embodiments, a different (greater or lesser) value for the angle $\theta_{sub.1}$ may be used. In still other embodiments, the entranceway could be configured to have a different shape.

[0084] The size of the throat **72** corresponds to the distance $T_{sub.1}$ between the inclined body surfaces **74** and **76** at the narrowest point of the entranceway **70**. The distance $T_{sub.1}$ is selected based on the size/diameter $D_{sub.1}$ of the arch wire **50** (or the range of arch wires) to be gripped by

the orthodontic gripping device **20**. The distance T.sub.1 must be large enough so that the arch wire **50** can be inserted into the arch wire receiving station **48** without the application of excessive force. But, care must be taken not to size the throat **72** so large as to permit the arch wire **50** to be released easily from the gripping jaw **52** through the throat **72**. It has been found that sizing the distance T.sub.1 to be within the range of between approximately 35% to approximately 40% of the diameter D.sub.1 tends to meet the above-described design objectives. In other embodiments, the distance T.sub.1 can be increased or decreased to suit a differently sized arch wire or a particular application.

[0085] The entranceway **70** (and more specifically, the throat **72**) opens onto (or provides access to) the arch wire receiving station **48**. In this embodiment, opposing, inner gripping surfaces **78** and **80** of the first and second jaw portions **44** and **46**, respectively, are configured with a concavity or indentation **82** which impart a generally circular profile to the arch wire receiving station **48**. Configured in this way, the profile of the arch wire receiving station **48** matches at least partially the cross-sectional (i.e. circular or substantially circular) shape of the arch wire **50**.

[0086] This arrangement tends to maximize contact/engagement between the inner gripping surfaces **78** and **80** and the outer surface **84** of the arch wire **50** and tends to improve retention of the arch wire **50** within the arch wire receiving station **48** by ensuring that a proper distribution of gripping forces on the arch wire is achieved. By matching the geometry of the arch wire receiving station **48** to that of the arch wire **50**, the risk of the arch wire being accidentally released from the gripping device in the event the arch wire or gripping device is exposed to a sudden impact, may be mitigated.

[0087] While matching the profile of the arch wire receiving station to that of the arch wire is generally preferred, it need not be the case in every application. In certain applications, the arch wire receiving station could be configured with a profile different from that of the arch wire.

[0088] Other modifications are possible. For instance, while in this embodiment the inner gripping surfaces **78** and **80** maintain the same arcuate profile throughout the length L.sub.1 of body **26**, in other embodiments the profiles of the inner gripping surfaces could be varied throughout the length of body. The inner gripping surfaces could feature an irregular surface profile. Instead of the inner gripping surfaces being smooth, they could be textured or roughened to enhance friction. Further still, the inner gripping surfaces could be coated with friction enhancing materials.

[0089] In this embodiment, by virtue of the circular profile of the arch wire receiving station **48**, the centre point for the radius of curvature of the inner gripping surface **78** coincides with the centre point for the radius of curvature of the inner gripping surface **80**, and both centre points lies on the longitudinal centreline CL.sub.1.

[0090] The arch wire receiving station **48** has a diameter D.sub.2 defined substantially by the opposing inner gripping surfaces **78** and **80**. The diameter D.sub.2 of the arch wire receiving station **48** is sized slightly smaller than the diameter D.sub.1 of the arch wire **50**. Preferably, the diameter D.sub.2 is sized 20% smaller than the diameter D.sub.1. In other embodiments, the sizing of diameter D.sub.2 may differ.

[0091] During the attachment procedure, when the arch wire **50** is received in the arch wire receiving station **48**, the first and second arm portions **40** and **42** deflect partially outwardly from each other to compensate for the fact that the diameter D.sub.2 of the arch wire receiving station **48** is smaller than the diameter D.sub.1 of the arch wire **50**. Because the orthodontic gripping device **20** is made from a superelastic metal alloy, the arm portions **40** and **42** are resilient and biased to their original (undeflected) positions. As the arm portions **40** and **42** seek to return to their original positions, their inner gripping surfaces **78** and **80**, which partially surround the arch wire **50**, exert a clamping or gripping force on the outer surface **84** of the arch wire **50** (as best shown in FIG. 4). The application of this gripping force tends to hold the arch wire **50** tightly and tends to resist displacement (or arrest movement) of the gripping device **20** relative to the arch wire **50**. It will thus be appreciated that the gripping jaw **52** derives its gripping force by putting to good advantage

the special spring-back characteristics of the metal alloy to tightly clamp onto the arch wire **50**. As a result, the gripping jaw **52** of the gripping device **20** tends to exert a greater gripping force on the arch wire than conventional arch wire stops and tends to offer better resistance against slippage. [0092] When the gripping device **20** is used as an arch wire stop, movement of the arch wire relative to the gripping device will preferably be resisted in the mesial/distal direction. However, it should be noted that the gripping device would discourage movement in all directions and would tend to resist rotation about the arch wire.

[0093] Opposite the throat **72**, the arch wire receiving station **48** opens onto a relatively narrow passageway **88**. The passageway **88** extends toward the base portion **38** and communicates with an arcuate (generally U-shaped) cutout **90** having opposed first and second portion **92** and **94**. The first portion **92** of the cutout **90** is defined between the first arm portion **40** and the base portion **38**, and follows an arcuate path from the passageway **88** toward the third corner **34** of the body **26**. The terminal end **96** of the first cutout portion **92** lines up roughly with the midpoint of the concavity **82** defined in the inner gripping surface **78** of the first gripping jaw portion **44**. Configured in this way, the first cutout portion **92** defines a peninsular section **98** in the first gripping jaw portion **44**.

[0094] In like fashion, the second portion **94** of the cutout **90** is defined between the second arm portion **42** and the base portion **38**, and follows an arcuate path from the passageway **88** toward the fourth corner **36** of the body **26**. The terminal end **100** of the second cutout portion **94** lines up roughly with the midpoint of the concavity **82** defined in the inner gripping surface **80** of the second gripping jaw portion **46**. Configured in this way, the second cutout portion **94** defines a peninsular section **102** in the second gripping jaw portion **46**. In other embodiments, cutout **90** could be configured differently.

[0095] The purpose of the cutout portions **92** and **94** is to permit the deflection of the gripping jaw portions **44** and **46** outwardly away from each other without the application of excessive force, when the arch wire **50** is being pushed through the throat **70** and into the arch wire receiving station **48**. The cutout portions **92** and **94** also allow the peninsular sections **98** and **102** of the gripping jaw portions **44** and **46** to deflect toward the first and second corners **30** and **32**, respectively, when the arch wire **50** is held in the arch wire receiving station **48** (as best shown in FIG. 4).

[0096] Turning now to FIGS. 3 and 4, a length of arch wire **50** is shown firmly retained by the gripping jaw **52** of the orthodontic gripping device **20**. As previously mentioned, the arch wire **50** in this embodiment has a circular or substantially circular cross-section and a diameter D.sub.1. It may be fabricated from linear elastic materials (e.g. stainless steel, cobalt chromium, titanium beta (III)), superelastic materials (including superelastic metal alloys, such as alloys of nickel titanium, nickel/titanium/copper, nickel/titanium/copper, nickel/titanium/copper/chromium), shape memory materials (including shape memory non-metallic materials, such as shape memory polymers; or shape memory metal alloys), or still other materials possessing superelastic or shape memory properties and suitable for use in orthodontic appliances.

[0097] For case of illustration, the arch wire **50** is shown conceptually as being a single solid strand of wire. However, the arch wire **50** can take the form of a plurality of braided wire strands or helically wrapped wire strands. An example of such a helically wrapped arch wire is the arch wire manufactured and sold by Strite Industries Limited of Cambridge, Ontario, Canada under the brand name SPEED Supercable™. The arch wire used in conjunction with the gripping device **20** can have a core or alternatively, it may be coreless (i.e. it may have a hollow cylindrical centre). An example of such a coreless arch wire is an arch wire manufactured and sold by Strite Industries Limited under the brand name SPEED Tubular Supercable™. In a further alternative, the arch wire can take the form of a Hills Dual Geometry Wire, that is, a wire with a first section provided with a square cross-section and second section provided with a circular cross-section. In such a case, the orthodontic gripping device **20** could be used on the second section of circular cross-section.

[0098] Referring to FIGS. 6, 7, 8 and 9, there is shown an orthodontic tool in the nature of a pair of pliers **110** which may be used to attach the gripping device **20** to the arch wire **50**. The pliers **110**

have first and second arms **112** and **114** arranged in cross-wise fashion and pivotally connected to each other by a pivot connector **116**. The first arm **112** includes a first end **118**, an opposed second end **120**, a first curved handle portion **122**, a first beak portion **124** and a first transition portion **126** for joining the first handle portion **122** to the first beak portion **124**. The first transition portion **126** is disposed at a location closer to the first end **118** than to the second end **120** and receives a portion of the pivot connector **116**. The first handle portion **122** extends from the second end **120** to the first transition portion **126**, while the first beak portion **124** runs from the first transition portion **126** to the first end **118**. The first beak portion **124** has a generally tapering profile, but terminates with a first work tip **128** that extends on a slant relative to the longitudinal axis P of the pliers **110**. In this embodiment, the angle α formed between the longitudinal axis P and the work tip **128** is 30 degrees (see FIG. 7). In other embodiments, a different angle α may be used.

[0099] The second arm **114** is generally structurally similar to the first arm **112** in that it too has a first end **130**, an opposed second end **132**, a second curved handle portion **134**, a second beak portion **136** and a second transition portion **138** joining the second handle portion **134** to the second beak portion **136**. The second transition portion **138** is disposed at a location closer to the first end **130** than to the second end **132** and receives another portion of the pivot connector **116**. Configured in this manner, the arms **112** and **114** are joined to each other by the pivot connector **116** at their respective transition portions **126** and **138**. The second handle portion **134** extends from the second end **132** to the second transition portion **138**, while the second beak portion **136** runs from the second transition portion **138** to the first end **130**. The second beak portion **136** also has a generally tapering profile (as best shown in FIG. 6), but terminates with a second work tip **140** that extends on the same slant relative to the longitudinal axis P of the pliers **110** as work tip **128**.

[0100] With reference to FIGS. 8 and 9, the first and second work tips **128** and **140** are now described in greater detail. The first work tip **128** has a generally block-like body **142** with an inner surface **144** oriented toward the second work tip **140**. Projecting from the inner surface **144** in a generally perpendicular orientation is a flange **146**. The flange **146** extends along the entire width of the body **142** and is carried close to the free end **148** of the work tip **128**. As will be explained in greater detail below, the flange **146** is configured for registration with a transverse slot **150** defined within the second work tip **140**. In an alternative embodiment, the flange could be configured differently.

[0101] Referring now specifically to FIG. 9, the second work tip **140** also includes a generally block-like body **152** with an inner surface **154** oriented toward the first work tip **128**. The inner surface **154** is formed with the first, generally C-shaped, transverse slot **150** which spans the width of the body **152**. The first transverse slot **150** is sized to accommodate a section of arch wire therein. In an alternative embodiment, the first transverse slot could be configured differently, for instance, it could be stepped so as to suit different sized arch wires.

[0102] A rectangular pocket or rebate **158** defined within the inner surface **154** partially overlies the transverse slot **150** with its short sides oriented generally parallel to the transverse slot **150**.

Advantageously, the rectangular rebate **158** is configured as a seat for holding an orthodontic gripping device, such as the gripping device **20**, during attachment of the gripping device to the arch wire. The provision of this seat tends to minimize the incidents of accidental drop or loss of the gripping devices **20** and facilitates handling of the gripping devices, which because of their relatively small size can be unwieldy. In other embodiments, the handling of the gripping devices could be improved by incorporating a spring-loaded mechanism into the rectangular rebate to securely retain the gripping device. Alternatively, one of the second work tip and the gripping device could be coated with a tacky material which would assist in holding the gripping device on the pliers **110**.

[0103] The body **152** is further provided with a second transverse slot **160** recessed from both the first transverse slot **150** and the rebate **158**. The second transverse slot **160** is co-extensive with, and generally parallel to, the short sides of the rebate **158**. The second transverse slot **160** provides

a clearance to accommodate any protrusions extending from the base portion **38** which may be left over from the gripping device manufacturing process. Additional slots may be provided for the same purpose.

[0104] The arms **112** and **114** of the pliers **110** are movable relative to each other between a closed setting (not shown) and an open setting **162** (best shown in FIG. **8**). In the closed setting, the flange **146** of the first male work tip **128** mates with the slot **150** of the second female work tip **140**. In the open setting **162**, a gap exists between the first and second work tips **128** and **140**. To move the arms **112** and **114** from the open setting **162** to the closed setting, the user brings the first and second handle portions **122** and **134** toward each other. The arms **112** and **114** are made to pivot about the pivot connector **116** causing the first and second beak portions **124** and **136** to move toward each other thereby closing the gap between the first and second work tips **128** and **140**. In an alternative embodiment, the pliers may be spring-loaded.

[0105] In a further embodiment, the arrangement of flange **146**, first and second transverse slots **150** and **160** and rectangular rebate **158** could be oriented differently on the first and second work tips **128** and **140**. For instance, in another embodiment, the arrangement could be rotated 90 degrees from the arrangement shown in FIG. **8**.

[0106] An exemplary procedure for attaching the orthodontic gripping device **20** to the arch wire **50** using the orthodontic pliers **110** is now explained in greater detail. The orthodontist moves the arms **112** and **114** of the pliers **110** to the open setting **162** and orients the pliers such that the second arm **114** is disposed above the first arm **112**. In this position, the rectangular rebate **158** defined in the second work tip **140** is accessible from the top. The orthodontist then places an orthodontic gripping device **20** into the second work tip **140** ensuring that the gripping device **20** is properly seated in the rectangular rebate **158** with its entranceway **70** oriented co-axially with the first transverse slot **150** and facing toward the first work tip **128**. The proper placement of the gripping device **20** within the rebate **158** is facilitated by the generally rectangular shape of the gripping device body **26** as the body **26** tends not to rotate in the rebate **158**. The gripping device **20** is now ready to be attached to the arch wire **50**.

[0107] The orthodontist can attach the gripping device **20** on the arch wire **50** either while the arch wire **50** is fitted on the patient's dental arch or prior to the placement the arch wire **50** in the orthodontic appliances or brackets mounted to the patient's teeth. The orthodontist positions the pliers **110** over the section of arch wire to be retained within the gripping device **20** and aligns that section of arch wire **50** with the entranceway **70** (defined in the gripping device body **26**) and the first transverse slot **150** of the second work tip **140**. Preferably, if the gripping device **20** is being attached to an arch wire **50** already fitted on the patient's upper dental arch, the entranceway **70** would be directed toward the gingival. On the other hand, if the gripping device **20** is being attached to an arch wire **50** already fitted on the patient's lower dental arch, preferably, the entranceway **70** would be directed toward the occlusal. This further helps to avoid the gripping device **20** accidentally falling out of the pliers **110** during the attachment procedure. This need not be the case in every application. In other embodiments, it may be possible to orient the entranceway **70** differently, for example, toward the lingual or toward the labial.

[0108] The orthodontist then squeezes the handle portions **122** and **134** together thereby causing the flange portion **146** of the first work tip **128** to act upon the section of arch wire **50** to be retained (as best shown in FIG. **10**). The flange portion **146** holds the arch wire **50** in place while the second work tip **140** causes the inclined body surfaces **74** and **76** to press against the arch wire. With the application of sufficient force, the arm portions **40** and **42** of the gripping device **20** will begin to deflect outwardly from each other and the throat **72** will widen to accommodate the arch wire **50** therein (see FIG. **11**). Once the arch wire **50** has cleared the throat **72**, the arm portions **40** and **42** will flex back toward each other in an attempt to return to their original (undeflected) positions. The inner gripping surfaces **78** and **80** of the first and second jaw portions **44** and **46** will come to bear on the outer surface **84** of the arch wire **50** and will hold the arch wire **50** tightly within the

gripping device **20** thereby discouraging displacement of the gripping device **20** relative to the arch wire **50**. With the gripping device **20** now attached to the arch wire **50**, the orthodontist can move the arms **112** and **114** of the pliers **110** to the open setting **162** and release the gripping device **20** from the tool.

[0109] It should be appreciated that the orthodontic pliers **110** are specifically designed for the gripping device **20** and other similar gripping devices. Accordingly, while it is generally preferred for ease and convenience that the pliers **110** be used to attach a gripping device **20** or other similar gripping device to an arch wire, this need not be the case in every application. Other tools could be employed to attach the gripping device **20** or other like gripping device to the arch wire.

[0110] In the embodiment shown and described, the gripping device **20** is made of nickel titanium, a metal alloy that behaves in a linear superelastic mode. It should be appreciated that the exemplary attachment procedure described above applies specifically to gripping devices made of metal alloys behaving in a superelastic mode. Where the gripping device is a material that behaves in a shape memory mode, the attachment procedure would differ as explained below.

[0111] As an initial step, the gripping device would be cooled to a temperature within a temperature range that allows significant deformation of the memorized shape of the shape memory material (referred to herein as the shape reset temperature range). Cooling may be achieved by placing the gripping device in contact with cooling device or cooling media such as dry ice, right up to the time in time when it is to be deployed. A similarly cooled tool would be then inserted into the entranceway of the gripping device body to cause the arms of the gripping device to deflect outward. Because the temperature of the gripping device is below the shape recovery temperature range of shape memory material (i.e. the temperature range within which the shape memory material recovers its memorized shape), the arms would remain in their respective flexed or deflected positions even after removal of the tool, and the entranceway would remain open large enough to allow an arch wire of the desired diameter or width to pass through it and into the jaw of the gripping device.

[0112] Thereafter the orthodontist would rapidly place the gripping device in the desired position over an arch wire installed in a patient's oral cavity or exterior to the patient's oral cavity (in both cases such that the arch wire is located within the jaws of the gripping device). The gripping device would then be heated up to a temperature falling within the shape recovery temperature range. As the gripping device reaches a temperature within the shape recovery temperature range, the shape memory properties in the material cause the arms to move back toward each other in an attempt to return to their original (undeflected) positions. The inner gripping surfaces of the first and second jaw portions come to bear on the outer surface of the arch wire and hold the arch wire tightly within the gripping device thereby discouraging displacement of the gripping device relative the arch wire.

[0113] In certain embodiments, it may be desirable to have a shape recovery temperature range that is below the temperature of the human oral cavity (which is on average 36.8° C. but may vary among individuals and time of day). For example, where the shape memory material is nickel titanium, a shape recovery temperature range of between -15° C. and 5° C. could be employed. Of course, other temperature ranges could also be used. In this way, shape memory of the gripping device may be obtained simply by allowing the gripping device to warm back up to the natural temperature in the oral cavity. During this time, it is recommended that the gripping device be held in position at the desired point of attachment on the arch wire to avoid sliding or shifting of the gripping device along the arch wire. Given the relatively small size of the gripping device, this should take a very short time. In alternative embodiments, the shape recovery temperature range could encompass the temperature of a patient's oral cavity. In such embodiments, the shape recovery temperature range could be, for example, between 17° C. and 37° C.

[0114] In still other embodiments, the shape recovery temperature range could be higher than the temperature of the human oral cavity (but less than the temperature at which permanent plastic

deformation occurs), for example, between 45° C. and 60° C. In such cases, a shape memory material having a shape reset temperature range that is below the temperature of the human oral cavity should be used, for example, between -36° C. and 0° C. Such a shape memory material should have a shape recovery temperature range sufficiently high such that the gripping device does not require constant cooling subsequent to being opened at low temperature. Moreover, the shape memory material would be selected to ensure that the gripping device retains sufficient gripping force when it cools to the temperature of the human oral cavity.

[0115] When the shape recovery temperature range is set higher than the temperature of the human oral cavity, the gripping device can be shipped and handled at ambient temperature and somewhat higher temperatures, including that of the human oral cavity, without causing the jaw of the gripping device to close. In such cases when attaching the gripping device to the arch wire, care should be taken to localize the application of the heat applied to the gripping device during the installation on an arch wire, and to take measures to prevent any thermal damage to the teeth or surrounding tissue near the point of installation.

[0116] Referring to FIGS. 5A, 5B and 5C, there is shown a second embodiment of an orthodontic gripping device constructed in accordance with the principles of the present invention. The gripping device in this embodiment is designated generally with reference numeral **170** and is similar in all material respects (i.e. functionality, configuration and construction) to the gripping device **20** shown in FIGS. **1** to **3**, except that gripping device **170** is provided with a hook **172** which depends from the gripping device body **174**.

[0117] The body **174** is similar to the body **26** in that it too includes a longitudinal centerline CL.sub.1 (shown in FIG. 5B), a transverse centerline CL.sub.2 (shown in FIG. 5C), a base portion **176** and a pair of opposed, spaced apart, first and second arm portions **178** and **180** connected to the base portion **176** and disposed on opposite sides of the centerline CL.sub.1. The body **174** is symmetrical about the longitudinal centerline CL.sub.1 such that the arm portions **178** and **180** are mirror images of each other. The body is also configured to be symmetrical about the transverse centerline CL.sub.2.

[0118] However, in contrast to body **26**, the body **174** has the hook **172** connected to the outer surface **182** of the base portion **176** and extending away from the first and second arm portions **178** and **180** along the centreline CL.sub.1. In this embodiment, the hook **172** is integrally formed with the body **174**. This need not be the case in every application. In other embodiments, the hook could be fabricated as a separate component and releasably or permanently fixed to the gripping device body.

[0119] Other modifications are also possible. For instance, the hook need not extend perpendicular to the base portion. It could extend from the base portion on a slant. While in this embodiment, the hook is centrally disposed on the base portion in alignment with the entranceway, in alternative embodiments, the hook could be positioned differently in relation to one of, or both, the base portion and the entranceway. In still other embodiments, the hook could be carried on one of the arm portions instead of on the base portion. The hook could extend from the arm portion in a perpendicular or slanted orientation. In embodiments where the hook is carried on one of the arm portions, it may be desirable to make the arm portion carrying the hook more robust (i.e. bigger and/or thicker) to better resist the forces applied to it by the hook. Conceivably, the hook could be mounted on any exterior surface of the gripping device body.

[0120] The hook **172** has a proximal end **184**, a free end **186** and a stem **188** which extends between the proximal and free ends **184** and **186**. At the proximal end **184**, the stem **188** flares outwardly on all sides with smoothly radiused surfaces. At a location closer to the free end **186** than to the proximal end **184**, the stem **188** again flares outwardly in all sides with smoothly radiused surfaces only to taper inwardly again a short distance later. This configuration defines the bulbous tip **190** of the hook **172**. In other embodiments, the hook **172** could be shaped differently.

[0121] In the embodiment shown in FIGS. 5A, 5B and 5C, the gripping device **170** is shown

provided with a hook **172**. In other embodiments, the gripping device could be outfitted with accessories other than a hook. For instance, in an alternative embodiment, an auxiliary slot or an eyelet could be formed in the body of the gripping device.

[0122] During orthodontic treatment, the gripping device **170** may serve a dual purpose-it can arrest movement of the arch wire in the mesial/distal direction while serving as an anchoring point for tying orthodontic elastic/elastomeric bands or ligature ties (metal or elastomeric) or traction force springs.

[0123] The gripping device **170** may be attached to the arch wire **50** in much the same manner as the gripping device **20**. However, the pliers **110** would have to be modified to provide a larger clearance than that currently provided by second transverse rebate **160** in order to accommodate the hook **172**.

[0124] In the embodiments shown in FIGS. **1** through **5C**, the gripping devices **20** and **170** are configured to receive and clamp tightly onto an arch wire of circular cross-section. However, this need not be the case in every application. In a third embodiment illustrated in FIGS. **12** to **17**, there is shown an orthodontic gripping device **200** which is configured for use with an arch wire **202** of substantially rectangular cross-section.

[0125] Referring to FIGS. **12**, **13A** and **13B**, the gripping device **200** is generally similar to gripping device **20** described above in that it too has a mesial face **204** and a distal face **206** and an elongate body **208** extending between the mesial and distal faces **204** and **206**. In this embodiment, the body **208** is of unitary (one-piece or monolithic) construction fabricated from a superelastic material. Preferably, such material is a superelastic metal alloy. Most preferably, the superelastic metal alloy is nickel titanium, or nickel titanium and other alloying elements (e.g. chromium (Cr), iron (Fe), vanadium (V), aluminium (Al), copper (Cu), cobalt (Co)), such as nickel/titanium/copper or nickel/titanium/copper/chromium. Preferably, the nickel titanium, or nickel titanium and other alloying elements, is cold worked. Although, cold worked and aged nickel titanium, or cold worked and aged nickel titanium and other alloying elements, could also be used. In this embodiment, the gripping device **200** is made of cold worked nickel titanium.

[0126] However, it should be appreciated that the gripping device could also be made from other materials suitable for use in orthodontic appliances and exhibiting spring-back properties similar to those exhibited by superelastic materials. For example, a gripping device fabricated from cold worked titanium beta III or solution heat treated and aged titanium beta III (a titanium molybdenum alloy) has been found to have superior gripping strength. Still other materials may be selected based on their ability to resist permanent deformation and/or mechanical failure when the gripping device is flexed during its engagement with the arch-wire, while at the same time possessing sufficient material stiffness, such that the gripping device exhibits sufficient spring-back to hold the arch-wire firmly in place once the gripping device is fully engaged with the arch wire.

[0127] In still other embodiments, the gripping device could be made of a shape memory material, such a shape memory metal alloy. Although, conceivably, a non-metallic material (such as a polymer) possessing shape memory properties could also be used.

[0128] Like body **26** of gripping device **20**, the body **208** has a vaguely rectangular shape or profile when seen in a mesial view such as shown in FIG. **13A**. The shape of the body **208** is defined substantially by four smoothly radiused corners-first corner **210**, second corner **212**, third corner **214** and fourth corner **216**. The corners **210**, **212**, **214** and **216** are rounded so as to avoid the formation of abrupt projecting edges which could otherwise irritate the soft tissue in the patient's mouth and cause discomfort when the orthodontic gripping device **200** is deployed in the patient's mouth.

[0129] The body **208** includes a longitudinal centreline CL.sub.1 (shown in FIG. **13A**), a transverse centerline CL.sub.2 (shown in FIG. **13B**), a base portion **218** and a pair of opposed, spaced apart, first and second arm portions **220** and **222** connected to the base portion **218** and disposed on opposite sides of the longitudinal centerline CL.sub.1. The body **208** is symmetrical about the

longitudinal centerline CL.sub.1 such that the arm portions **220** and **222** are mirror images of each other. The body **208** is also configured to be symmetrical about a transverse centerline CL.sub.2. In other embodiments, the body could be configured asymmetrically about one or both the longitudinal centerline CL.sub.1 and the transverse centerline CL.sub.2.

[0130] The first arm portion **220** joins the base portion **218** at the first corner **210**, while the second arm portion **222** attaches to the base portion **218** at the second corner **212**. Each arm portion **220**, **222** extends away from corner **210**, **212**, respectively, in a direction generally perpendicular to the base portion **218**. At the third corner **214**, the first arm portion **220** turns inwardly toward the second arm portion **222** and ultimately, terminates with a first stub-like projection **224**. The section of the first arm portion **220** which runs between the first stub-like projection **224** and the base portion **218** defines a first jaw portion **226**. Similarly, the second arm portion **222** turns inwardly toward the first arm portion **220** at the fourth corner **216** to terminate with a second stub like projection **228**. The section of the second arm portion **222** which runs between the second stub-like projection **228** and the base portion **218** defines a second jaw portion **230**.

[0131] The first and second jaw portions **226** and **230** are arranged opposite one another and separated by a relatively small gap which defines an arch wire receiving slot or station **232** for captively engaging (securely retaining) the arch wire **202** therein. Configured in this manner, the first and second jaw portions **226** and **230** together define the gripping jaw **234** of the orthodontic gripping device **200** (as best shown in FIG. 12).

[0132] With reference to FIGS. 12, 13A and 13B, the elongate body **208** further includes a length L.sub.2 measured between the mesial and distal faces **204** and **206**, a width W.sub.2 measured between the outer lateral surface **240** of the first arm portion **220** and the outer lateral surface **246** of the second arm portion **222**, and a depth H.sub.2 measured between the outer surface **248** of the base portion **218** and the outer surfaces **250** of the arm portions **220** and **222** disposed opposite to the outer surface **248**. In this embodiment, the body **208** is nearly as long as it is wide. In other embodiments, the body may be configured to have a length L.sub.2 equal to, lesser than or greater than the width W.sub.2.

[0133] Roughly midway between the third and fourth corners **214** and **216** and opposite the base portion **218**, an arch wire entranceway **252** is formed within the body **208**. The entranceway **252** runs the entire length of the elongate body **208** and is generally funnel-shaped with a taper in the direction of the base portion **218** ultimately culminating in a throat **254**. The funnel shape of the entranceway **252** is defined by a first inclined surface **256** presented by the first stub-like projection **224** and a second inclined surface **258** presented by the second stub-like projection **228**. The inclined body surfaces **256** and **258** are separated from each other by an angle θ .sub.2 (not represented in the figures). In this embodiment, the angle θ measures approximately 24 degrees. It will be appreciated that in other embodiments, a different (greater or lesser) value for the angle θ .sub.2 may be used.

[0134] The size of the throat **254** corresponds to the distance T.sub.2 between the inclined surfaces **256** and **258** at the narrowest point of the entranceway **252**. The distance T.sub.2 is selected based on the size (i.e. width W.sub.3) of the arch wire **202** (or the range of arch wires) to be retained by the orthodontic gripping device **200**. The distance T.sub.2 must be large enough so that the gripping device **200** can be fitted onto the arch wire **202** without the application of excessive force. But, care must be taken not to size the throat **254** so large as to permit the arch wire **202** to be released easily from the gripping jaw **234** through the throat **254**. It has been found that sizing the distance T.sub.2 to be within the range of between approximately 65% to approximately 70% of the width W.sub.3 tends to meet the above-described design objectives. In other embodiments, the distance T.sub.2 can be increased or decreased to suit a differently sized arch wire or a particular application.

[0135] The entranceway **252** (and more specifically, the throat **254**) opens onto (or provides access to) the arch wire receiving station **232**. In this embodiment, the arch wire receiving station **232** is shaped generally like a quatrefoil (but for the missing fourth side corresponding to the location of

the throat **254**). Defining this quatrefoil-like shape are four rounded undercuts or clearances formed in the body **208**—a first undercut **260** located adjacent the first corner **210**, a second undercut **262** located adjacent the second corner **212**, a third undercut **264** located adjacent the third corner **214** and a fourth undercut **266** located adjacent the fourth corner **216**—and three inner surfaces—a pair of opposing, inner gripping surfaces **268** and **270** of the first and second jaw portions **226** and **230** and an inner surface **272** of the base portion **218** as best shown in FIG. **13A**. The undercuts **260**, **262**, **264** and **266** are provided to accommodate the radiused corners **274**, **276**, **278** and **280** of the arch wire **202**. While not generally preferred, in alternative embodiments, the arch wire receiving station could be configured without any undercuts. For example, the arch wire receiving station could have a generally square or rectangular profile.

[0136] The inner surface **272** defines a seat for supporting side **288** of the arch wire **202**. It extends between the first and second undercuts **260** and **262** and is substantially parallel with the outer surface **248** of the base portion **218**.

[0137] The inner gripping surface **268** of the first jaw portion **226** extends between the first and third undercuts **260** and **264**, while the inner gripping surface **270** of the second jaw portion **230** runs between the second and fourth undercuts **262** and **266**. Each of the inner gripping surfaces **268** and **270** has a slightly skewed orientation relative to the longitudinal centerline CL.sub.1. More specifically, as the inner gripping surface **268** extends from the first undercut **260** to the third undercut **264** it tapers inwardly toward the opposed gripping surface **270**. Similarly, the inner gripping surface **270** tapers inwardly toward the gripping surface **268** in the direction of the fourth undercut **266**. Stated differently, the first and second inner gripping surfaces **268** and **270** can be said to converge toward each other in the direction of the entranceway **252**. In this embodiment, the angle of taper θ .sub.3 for each gripping surface **268**, **270** (as measured between a vertical axis V (generally parallel to the longitudinal centerline CL.sub.1) and each inner gripping surface **268**, **270**, as the case may be) is 3 degrees.

[0138] This taper tends to encourage better contact/engagement between the gripping surfaces **268** and **270** and the lateral sides **284** and **286** of the arch wire **202** when the arch wire **202** is received within the arch wire receiving station **232** and the arm portions **220** and **222** are partially deflected. This tends to improve retention of the arch wire **202** within the arch wire receiving station **232** by ensuring that a proper distribution of gripping forces on the arch wire is achieved. In this way, the risk of the arch wire being accidentally released from the gripping device in the event the arch wire or gripping device is exposed to a sudden impact, may be mitigated.

[0139] In other embodiments, a different (i.e. greater or lesser) angle of taper θ .sub.3 for both inner gripping surfaces may be used. In a further alternative, these inner gripping surfaces could be provided with no taper whatsoever. In such an embodiment, the first and second inner gripping surfaces would be oriented substantially parallel to the longitudinal centerline CL.sub.1.

[0140] Preferably, the width of the arch wire receiving station **232** at its narrowest point (as measured between the inner gripping surfaces **268** and **270** at the location where these surfaces meet with the third undercut **264** and the fourth undercut **266**, respectively) is at least 10% smaller than the width W.sub.3 of the arch wire **202**. However, in other embodiments, the width of the arch wire receiving station could be sized differently.

[0141] Other modifications are possible. For instance, while in this embodiment the inner gripping surfaces **268** and **270** maintain the same profile throughout the length L.sub.2 of body **208**, in other embodiments the profiles of the inner gripping surfaces could be varied throughout the length of body. The inner gripping surfaces could be irregular. Instead of the inner gripping surfaces being smooth, they could be textured or roughened to enhance friction. Further still, the inner gripping surfaces could be coated with friction enhancing materials.

[0142] In this embodiment, because the inner gripping surfaces **268** and **270** have a skewed orientation, the gap or distance G between the surfaces **268** and **270** is not constant. The gap G at the narrowest point between the inner gripping surfaces **268** and **270** (i.e. where the gripping

surfaces **268** meets the third undercut **264** and the gripping surface **270** meets the fourth undercut **266**) is sized smaller than the width $W_{sub.3}$. In embodiments where the inner gripping surfaces are oriented substantially parallel to the longitudinal centerline $CL_{sub.1}$, the gap G as measured at any location between the inner gripping surfaces will also be sized smaller than the width $W_{sub.3}$. [0143] During the attachment procedure, when the arch wire **202** is received in the arch wire receiving station **232**, the first and second arm portions **220** and **222** deflect partially outwardly from each other to compensate for the fact that the gap G measured at the narrowest point between the inner gripping surfaces **268** and **270** is smaller than the width $W_{sub.3}$ of the arch wire **202**. Because the orthodontic gripping device **200** is made from a superelastic metal alloy, the arm portions **220** and **222** are resilient and biased to their original (undeflected) positions. As the arm portions **220** and **222** seek to return to their original positions, their inner gripping surfaces **268** and **270** apply a clamping or gripping force on the lateral sides **284** and **286** of the arch wire **202** (as best shown in FIG. 15). The application of this gripping force tends to hold the arch wire **202** tightly and tends to resist displacement (or arrest movement) of the gripping device **200** relative to the arch wire **202**. It will thus be appreciated that the gripping jaw **234** derives its gripping force by putting to good advantage the special spring-back characteristics of the metal alloy to clamp tightly onto the arch wire **202**. As a result, the gripping jaw **234** of the gripping device **200** tends to exert a greater gripping force on the arch wire than conventional arch wire stops and tends to offer better resistance against slippage.

[0144] When the gripping device **200** is used as an arch wire stop, displacement will preferably be resisted in the mesial/distal direction. However, it should be noted that the gripping device would also discourage movement in all directions and also tends to resist rotation about the arch wire.

[0145] Turning now to FIGS. 14 and 15, a length of arch wire **202** is shown firmly retained by the gripping jaw **234** of the orthodontic gripping device **200**. The arch wire **202** in this embodiment has a rectangular or substantially rectangular cross-section defined by lateral sides **284** and **286** (i.e. the short sides) extending between rounded corners **274** and **278**, and rounded corners **276** and **280**, respectively, and opposed sides **282** and **288** (i.e. the long sides) running between rounded corners **278** and **280**, and rounded corners **274** and **276**. The width $W_{sub.3}$ of the arch wire **202** is defined as the distance between the lateral sides **284** and **286**, while the depth $H_{sub.3}$ of the arch wire **202** is defined by the distance between the sides **282** and **288**.

[0146] For case of illustration, the arch wire **202** is shown conceptually as being a single solid strand of wire. However, the arch wire **202** can take the form of a plurality of braided wire strands or helically wrapped wire strands. This multi-stranded arch wire may be formed (e.g. by rolling) to have a generally rectangular cross-section with rounded corners and may possess a core or be coreless.

[0147] Attachment of the orthodontic gripping device **200** to the arch wire **202** is similar to the attachment of the gripping device **20** to arch wire **50** described above. The orthodontist moves the arms **112** and **114** of the pliers **110** to the open setting **162** and orients the pliers such that second arm **114** is disposed above the first arm **112**. In this position, the rectangular rebate **158** defined in the second work tip **140** is accessible from the top. The orthodontist then places an orthodontic gripping device **200** into the second work tip **140** ensuring that the gripping device **200** is properly seated in the rectangular rebate **158** with its entranceway **252** oriented parallel to the first transverse slot **150**. The gripping device **200** is now ready to be attached to the arch wire **202**.

[0148] The orthodontist can attach the gripping device **200** on the arch wire **202** either while the arch wire **202** is fitted on the patient's dental arch or prior to the placement of the arch wire **202** in the orthodontic appliances or brackets mounted to the patient's teeth. The orthodontist positions the pliers **110** over the section of arch wire to be retained within the gripping device **200** and aligns that section of arch wire with the entranceway **252** (defined in the gripping device body **208**) and the first transverse slot **150** of the second work tip **140**. Preferably, if the gripping device **200** is being attached to an arch wire **202** already fitted on the patient's upper dental arch, the entranceway **252**

will be directed toward the gingival. On the other hand, if the gripping device **200** is being attached to an arch wire **202** already fitted on the patient's lower dental arch, preferably, the entranceway **252** will be directed toward the occlusal. This further helps prevent the gripping device **200** from accidentally falling out of the pliers **110** during the attachment procedure. This need not be the case in every application. In other embodiments, it may be possible to orient the entranceway **252** differently, for example, toward the lingual or toward the labial.

[0149] The orthodontist then squeezes the handle portions **122** and **134** together thereby causing the flange portion **146** of the first work tip **128** to act upon the section of arch wire **202** to be retained (as best shown in FIG. **16**). The flange portion **146** holds the arch wire **202** in place while the second work tip **140** causes the inclined body surfaces **256** and **258** to press against the arch wire. With the application of sufficient force, the arm portions **220** and **222** of the gripping device **200** will begin to deflect outwardly from each other and the throat **254** will widen to accommodate the arch wire **202** therein (see FIG. **17**). Once the arch wire **202** has cleared the throat **254**, the arm portions **220** and **222** will flex back toward each other in an attempt to return to their original (undeflected) positions. The inner gripping surfaces **268** and **270** of the first and second jaw portions **226** and **230** will come to bear on the lateral faces **284** and **286** of the arch wire **202** and will hold the arch wire **202** tightly within the gripping device **200** thereby discouraging displacement of the arch wire **200** in any direction. With the gripping device **200** now attached to the arch wire **202**, the orthodontist can move the arms **112** and **114** of the pliers **110** to the open setting **162** and release the gripping device **200** from the tool.

[0150] In the embodiment shown and described, the gripping device **200** is made of nickel titanium, a metal alloy that behaves in a linear superelastic mode. Where the gripping device is a material that behaves in a shape memory mode, the attachment procedure would be similar to that described above at paragraphs [00109] to [00113].

[0151] It will be appreciated that the gripping device **200** could be used with an arch wire having a rectangular cross-section defined by right angle corners. With appropriate modifications the gripping device **200** could also be used with an arch wire of square cross-section (defined by right angle corners or rounded corners). Further still, the principles of the present invention could be applied to design a gripping device suitable for use with arch wires having cross-sections other than circular, rectangular or square. For instance, a gripping device could be configured to clamp onto an arch wire having a generally D-shaped profile (such as, the SPEED D-Wire™ manufactured and sold by Strite Industries Limited of Cambridge, Ontario, Canada), or a profile comprising two right angle corners and one radiused corner (such as, the SPEED Wire™ manufactured and sold by Strite Industries Limited of Cambridge, Ontario, Canada).

[0152] In such cases, a gripping device would be made of a superelastic material or shape memory material, beta-titanium III or another suitable material and could be structurally similar to the gripping devices **20** and **200** with an arrangement of arm portions and a base portion. Each arm portion could include a jaw portion, which together with the other jaw portion could form a gripping jaw. The jaw portions could be spaced apart from each other with the gap extending between them sized smaller than the lateral dimension or width of the arch wire. Moreover, the jaw portions could be provided with inner gripping surfaces configured to engage surfaces of the arch wire. In some cases, the profiles of such inner gripping surfaces could be designed to match closely (or correspond to) the shape of the arch wire to be gripped.

[0153] Other modifications are possible. In other embodiments, the gripping devices could be provided with means for facilitating handling of the gripping device to better orient or position the gripping device within a pair of pliers or other tool. In one embodiment, such means could take the form of grooves, notches or indents registerable with mating projections on a pair of pliers or other tool. Examples of such embodiments are shown in FIGS. **18** to **21**.

[0154] FIGS. **18** and **19** depict a gripping device **300** generally similar to gripping device **20** described above in all material respects except that the that gripping device **300** includes a pair of

opposed, first and second, elongate grooves or notches **302** and **304** formed on the gripping device body **306**. The first groove **302** is defined on the outer surface **308** of the first arm portion **310** and runs the entire length of the body **306**. In this embodiment, the first groove **302** is disposed between the first and second corners **312** and **314** of the body **306**, closer to second corner **314**. Similarly, the second groove **304** is defined on the outer surface **316** of the second arm portion **318** and also runs the entire length of the body **306**. The second groove **304** is disposed between the third and fourth corners **320** and **322** of the body **306**, closer to fourth corner **322**. In this embodiment, the grooves **302** and **304** have an arcuate profile. In other embodiments, the grooves could have a different profile and be arranged differently. For instance, in other embodiments, the grooves could be made to extend only partway along the body. The grooves could be formed closer to the base portion. Alternatively, the outer surface of each arm portion could be provided with more than one groove. In still other embodiments, the means for facilitating handling of the gripping device could take a different form. For example, instead of providing grooves on the gripping device body which are registerable with projections on a tool, the gripping device body could be provided with projections which mate with grooves formed on the work tip surface of the tool.

[0155] FIGS. **20** and **21** show a gripping device **320** generally similar to gripping device **200** described above in all material respects except that the gripping device **320** includes a pair of opposed, first and second, elongate grooves or notches **322** and **324** formed on the gripping device body **326**. The configuration of the grooves **322** and **324** and their arrangement on the body **326** is generally similar to the configuration of grooves **302** and **304** and their arrangement on body **306** such that no further description is required.

[0156] FIGS. **22** and **23** show a gripping device **340** in accordance with a sixth embodiment of the present invention. The gripping device **340** is generally similar to gripping device **300** described above in all material respects except that the gripping device **340** has an arch wire receiving station **342** that is configured differently than the arch wire receiving station of gripping device **300** and the gripping device **340** includes a different body **343** with a somewhat reinforced (or thicker) base portion **344** than the base portion of the gripping device **300**.

[0157] More specifically, in contrast to the arch wire receiving station of the gripping device **300** which has a circular profile (not unlike arch wire receiving station **48** shown in FIG. **2A**), the arch wire receiving station **342** has an oval or elliptical profile (demarcated by dashed lines in FIG. **23**). The oval profile is defined substantially by opposed inner gripping surfaces **346** and **348** of the first and second jaw portions **350** and **352**, respectively. In this embodiment, the major axis of the oval profile is aligned with the longitudinal centerline CL.sub.1. In other embodiments, the major axis could be disposed parallel to, but offset from, the longitudinal centerline CL.sub.1.

[0158] By virtue of the oval profile of the arch wire receiving station **342**, the centre point for the radius of curvature of the inner gripping surface **346** is offset from the centre point for the radius of curvature of the inner gripping surface **348**, and both centre points are offset from the longitudinal centreline CL.sub.1.

[0159] The arch wire receiving station **342** has a short diameter D.sub.s (i.e. the diameter measured along its minor axis) defined by the opposing inner gripping surfaces **346** and **348**. The diameter D.sub.s of the arch wire receiving station **342** is sized slightly smaller than the diameter D.sub.1 of the arch wire **50**. Preferably, the diameter D.sub.s is sized between about 10% to about 15% smaller than the diameter D.sub.1. In other embodiments, the sizing of diameter D.sub.s may differ.

[0160] It has been found that by configuring the arch wire receiving station **342** with an oval profile enhanced gripping forces may be generated when an arch wire of circular or substantially circular cross-section is placed in the gripping device **340**.

[0161] In this embodiment, the base portion **344** is more prominent than the base portion of the gripping device **300** in order to support a pair of first and second, lateral wing members (or deflectable fingers) **354** and **356** which depend from, and are integrally formed with, the base portion **344**. The first lateral wing member **354** extends from the corner portion **358** of the gripping

device body **343** substantially in the direction of the arch wire receiving station **342** in a curved, finger-like fashion. The first lateral wing member **354** terminates at a free end **360** which stops shy of an elongate groove or notch **362** defined in the gripping device body **343** so as not obstruct access thereto. The notch **362** is similar in structure and purpose to notch **302** formed in gripping device body **306**. Defined between the first lateral wing member **354** and the first arm portion **355** of the gripping device body **343** is an elongate gap **364**.

[0162] Similarly, the lateral wing member **356** extends from the corner portion **366** of the gripping device body **343** substantially in the direction of the arch wire receiving station **342** in a curved, finger-like fashion. The lateral wing member **356** terminates at a free end **368** which stops shy of an elongate groove or notch **370** defined in the gripping device body **343** so as not obstruct access thereto. The notch **370** is similar in structure and purpose to notch **304** formed in gripping device body **306**. An elongate gap **372**, similar to gap **364**, extends between the lateral wing member **356** and the second arm portion **374** of the gripping device body **343**. As will be explained in greater detail below, the gaps **364** and **372** provide a clearance for the lateral wing members **354** and **356** to allow them to move between their respective undeflected positions **380** and their respective deflected positions **382**.

[0163] In other embodiments, the lateral wing members could be configured differently.

[0164] The purpose of the lateral wing members **354** and **356** is to facilitate handling of the gripping device **340** by an orthodontist using the orthodontic pliers **110** described earlier (or the like). More specifically and as shown in FIGS. **24A** and **24B**, the lateral wing members **354** and **356** serve as resilient stops for preventing the accidental release of the gripping device **340** when it is seated within the rectangular rebate **158** formed in the second work tip **140** of the pliers **110**.

[0165] When the lateral wing members **354** and **356** are in their respective undeflected positions **380** (shown in FIGS. **23** and **24A**), the gripping device **340** has a width $W_{sub.5}$ (as measured between the outer lateral surface **384** of the first lateral wing member **354** and the outer lateral surface **386** of the second lateral wing member **356**) that is larger than the width of the rectangular rebate **158**. Conversely, when a force is applied on the gripping device **340** (either manually by the orthodontist using his/her finger, or mechanically, using the flange **146** formed on the first work of the pliers **110**), the lateral wing members **354** and **356** are urged to move to their respective deflected positions **382** (shown in FIGS. **24B**), wherein the free ends **360** and **368** encroach onto the gaps **364** and **372** such that the width $W_{sub.5}$ is slightly smaller than the width of the rectangular rebate **158**. This allows the gripping device **340** to be accommodated and captively retained within the rebate **158**.

[0166] Because the orthodontic gripping device **340** is made from a superelastic metal alloy, the lateral wing members **354** and **356** are resilient and biased to their original (undeflected) positions **380**. As lateral wing members **354** and **356** seek to return to their original positions, the outer lateral surfaces **384** and **386** of the lateral wing members **354** and **356** are urged against the walls **388** and **390** which define the rectangular rebate **158**, exerting an outward force thereupon. It is these forces which operate to retain the gripping device within the rectangular rebate **158**.

[0167] The attachment of the orthodontic gripping device **340** to an arch wire is generally similar to the attachment of the gripping device **20** to arch wire **50** described above such that no further description is required.

[0168] FIGS. **25** and **26** show a gripping device **400** generally similar to gripping device **320** described above in all material respects except that the gripping device **400** includes a different body **402** with a somewhat reinforced (or thicker) base portion **404** from which depend a pair of first and second lateral wing members **406** and **408**. In this regard, the structure of the base portion **404** and the configuration and arrangement of the lateral wing members **406** and **408** generally resembles those of the base portion **344** and the lateral wing members **354** and **356** such that the description of the latter elements will suffice for the former elements.

[0169] It will be appreciated that the gripping devices shown in FIGS. **2A** and **13A** could also be

constructed with thickened base portions and lateral wing members as described above.

[0170] In the embodiments shown in FIGS. 1 to 5 and 12 to 15, 18 to 23, 25 and 26, the gripping devices all have a split-body configuration particularized by a funnel-shaped entranceway leading into the body and providing access to the arch wire receiving station. In other embodiments, the gripping device body could be shaped differently. In one alternative embodiment, the gripping device body could have an annular or donut shaped body with spaced apart, deflectable arm portions (similar to those described above in the context of the other embodiments shown in the figures) contained within the envelope of the annular body. In such cases, the entranceway would be defined by the apertures defined in the mesial and distal faces of the body giving access to the arch wire receiving station. The gripping device could be placed onto the arch wire by urging the tip of the arch wire through the arch wire receiving station.

[0171] Any of the arch wire gripping devices described above or other arch wire gripping devices constructed in accordance with the principles of the present invention could be provided with accessories in the form of a hook (such as hook 172, shown in FIG. 5A), an auxiliary slot, an eyelet, or the like.

[0172] Any of the arch wire gripping devices described above or other arch wire gripping devices constructed in accordance with the principles of the present invention could be used as end stops (i.e. to restrict displacement at the ends of the arch wire). Alternatively, the gripping devices could be employed to resist displacement of a particular bracket/tooth arrangement relative to the arch wire.

[0173] The foregoing disclosure has described exemplary uses of gripping devices with arch wires. While gripping devices made in accordance with the principles of the present invention tend to be well-suited for use in conjunction with arch wires, it should be appreciated that use of the gripping device is not limited to arch wires. Gripping devices constructed in accordance with the principles of the present invention could be employed to similar advantage in conjunction with wires other than arch wires or further still, with auxiliary dental devices or other dental components needing to be gripped or tightly retained. Moreover, the principles of the present invention need not be restricted solely to dental/orthodontic applications. With appropriate modifications, the gripping devices described above could be adapted for gripping objects other than arch wires or dental components.

[0174] Although the foregoing description and accompanying drawings relate to specific preferred embodiments of the present invention as presently contemplated by the inventor, it will be understood that various changes, modifications and adaptations, may be made without departing from the spirit of the invention.

Claims

1. An orthodontic gripping device for attaching to an arch wire, the gripping device comprising: a body made of a material selected from the group consisting of a superelastic metal alloy, cold worked titanium beta III, and solution heat treated and aged titanium beta III; the body having first and second, spaced apart arm portions connected to each other; the first arm portion including a first jaw portion having a first inner, arch wire-gripping surface; the second arm portion including a second jaw portion having a second inner, arch wire-gripping surface; the first and second inner gripping surfaces being disposed opposite one another in spaced relation, the space between the first and second inner gripping surfaces defining a station for receiving the arch wire therein, at least a portion of the arch wire receiving station being sized smaller than the arch wire; the body further including an entranceway defined between the first and second arm portions, the entranceway providing access to the arch wire receiving station; the first and second arm portions being resiliently deflectable outwardly away from each other when the arch wire is admitted through the entranceway into the arch wire receiving station; the first and second inner gripping

surfaces engaging the arch wire and applying opposing gripping forces against the arch wire so as to resist displacement of the gripping device relative to the arch wire, when the arch wire is seated within the arch wire receiving station.

2. The gripping device of claim 1 wherein the body is of unitary construction.
3. The gripping device of claim 1 wherein at least one of the arm portions is a separate component.
4. The gripping device of claim 1 wherein the superelastic metal alloy is selected from the group consisting of: (a) cold worked nickel titanium; (b) cold worked and aged nickel titanium; (c) cold worked nickel titanium and other alloying elements; and (d) cold worked and aged nickel titanium and other alloying elements.
5. The gripping device of claim 1 wherein the superelastic metal alloy behaves in a linear superelastic mode.
6. The gripping device of claim 1 wherein the superelastic metal alloy behaves in a non-linear superelastic mode.
7. The gripping device of claim 1 wherein the body is made of a material consisting of: (a) cold worked titanium beta III; and (b) solution heat treated and aged titanium beta III.
8. The gripping device of claim 1 wherein: the body has a longitudinal centerline; the first and second arm portions are disposed on opposite sides of the longitudinal centerline; and the first and second arm portions are mirror images of each other.
9. The gripping device of claim 1 wherein the body has a longitudinal centerline and the body is symmetrical about the longitudinal centerline.
10. The gripping device of claim 1 wherein the body has a longitudinal centerline and the body is asymmetrical about the longitudinal centerline.
11. The gripping device of claim 1 wherein the body has a transverse centerline and the body is symmetrical about the transverse centerline.
12. The gripping device of claim 1 wherein the body has a transverse centerline and the body is asymmetrical about the transverse centerline.
13. The gripping device of claim 1 wherein the body further includes a base portion extending between the first and second arm portions to connect one to the other.
14. The gripping device of claim 13 wherein the entranceway is disposed opposite the base portion.
15. The gripping device of claim 1 wherein: the arch wire has a profile defined by an outer surface; and the first and second gripping surfaces are configured to substantially conform to at least a portion of the arch wire's outer surface.
16. The gripping device of claim 15 wherein: the profile of the arch wire is generally circular; and each of the first and second inner gripping surfaces has an arcuate profile configured to match at least partially the curvature of the arch wire's profile.
17. The gripping device of claim 1 wherein: the arch wire has a diameter; the arch wire receiving station has a substantially circular profile and a diameter; and the diameter of the arch wire receiving station is sized smaller than the diameter of the arch wire.
18. The gripping device of claim 1 wherein: the arch wire has a width; and at least a portion of the arch wire receiving station is sized smaller than the width of the arch wire.
19. The gripping device of claim 18 wherein: the arch wire has a width; the arch wire receiving station has a width of constant dimension; and the width of the arch wire receiving station is sized smaller than the width of the arch wire.
20. The gripping device of claim 18 wherein: the body has a longitudinal centerline; the first and second arm portions are disposed on opposite sides of the longitudinal centerline; the first and second arm portions are mirror images of each other, and the first and second inner gripping surfaces are oriented substantially parallel to the longitudinal centerline.
